In [3]:
```
import pandas as pd
```

In [7]:
```
df = pd.read_excel(r"C:\Users\Suresh Kunala\Downloads\data.xlsx")
```

In [8]:
```
df
```

Out[8]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [9]:
```
df['CountryCode']
```

Out[9]:
```
0      ABW
1      AFG
2      AGO
3      ALB
4      ARE
      ...
190    YEM
191    ZAF
192    COD
193    ZMB
194    ZWE
Name: CountryCode, Length: 195, dtype: object
```

In [11]:
```
df['InternetUsers']
```

Out[11]:
```
0      78.9
1       5.9
2      19.1
3      57.2
4      88.0
       ...
190    20.0
191    46.5
192     2.2
193    15.4
194    18.5
Name: InternetUsers, Length: 195, dtype: float64
```

In [13]:
```
df[['CountryName','BirthRate']]
```

Out[13]:

	CountryName	BirthRate
0	Aruba	10.244
1	Afghanistan	35.253
2	Angola	45.985
3	Albania	12.877
4	United Arab Emirates	11.044
...	...	...
190	Yemen, Rep.	32.947
191	South Africa	20.850
192	Congo, Dem. Rep.	42.394
193	Zambia	40.471
194	Zimbabwe	35.715

195 rows × 2 columns

In [15]:

```
df_cat = df[['CountryName','BirthRate']]
df_cat
```

Out[15]:

	CountryName	BirthRate
0	Aruba	10.244
1	Afghanistan	35.253
2	Angola	45.985
3	Albania	12.877
4	United Arab Emirates	11.044
...	...	...
190	Yemen, Rep.	32.947
191	South Africa	20.850
192	Congo, Dem. Rep.	42.394
193	Zambia	40.471
194	Zimbabwe	35.715

195 rows × 2 columns

In [16]:
```
df.shape
```

Out[16]:
```
(195, 5)
```

In [17]:
```
df_cat.shape
```

Out[17]:
```
(195, 2)
```

In [18]:
```
df.columns
```

Out[18]:
```
Index(['CountryName', 'CountryCode', 'BirthRate', 'InternetUsers',
       'IncomeGroup'],
      dtype='object')
```

In [20]:
```
df_cat.columns
```

Out[20]: Index(['CountryName', 'BirthRate'], dtype='object')

In [24]:
```
df.describe()
```

Out[24]:

	BirthRate	InternetUsers
count	195.000000	195.000000
mean	21.469928	42.076471
std	10.605467	29.030788
min	7.900000	0.900000
25%	12.120500	14.520000
50%	19.680000	41.000000
75%	29.759500	66.225000
max	49.661000	96.546800

In [25]:
```
df_cat.describe()
```

Out[25]:

	BirthRate
count	195.000000
mean	21.469928
std	10.605467
min	7.900000
25%	12.120500
50%	19.680000
75%	29.759500
max	49.661000

In [26]:
```
df
```

Out[26]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
...	...	...	...	...	...
190	Yemen, Rep.	YEM	32.947	20.0	Lower middle income
191	South Africa	ZAF	20.850	46.5	Upper middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

195 rows × 5 columns

In [30]:
```
df['BirthRate'] <20
```

Out[30]:
```
0       True
1      False
2      False
3       True
4       True
       ...
190    False
191    False
192    False
193    False
194    False
Name: BirthRate, Length: 195, dtype: bool
```

In [32]:
```
df[df['BirthRate'] <20]
```

Out[32]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income
5	Argentina	ARG	17.716	59.9	High income
6	Armenia	ARM	13.308	41.9	Lower middle income
...	...	...	...	...	...
181	United States	USA	12.500	84.2	High income
183	St. Vincent and the Grenadines	VCT	16.306	52.0	Upper middle income
184	Venezuela, RB	VEN	19.842	54.9	High income
185	Virgin Islands (U.S.)	VIR	10.700	45.3	High income
186	Vietnam	VNM	15.537	43.9	Lower middle income

100 rows × 5 columns

In [35]:
```
df[df['CountryName']== 'Afghanistan']
```

Out[35]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
1	Afghanistan	AFG	35.253	5.9	Low income

In [38]:
```
df[(df.BirthRate > 10) & (df.InternetUsers< 20)]
```

Out[38]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
11	Burundi	BDI	44.151	1.3	Low income
13	Benin	BEN	36.440	4.9	Low income
14	Burkina Faso	BFA	40.551	9.1	Low income
...	...	...	...	...	...
187	Vanuatu	VUT	26.739	11.3	Lower middle income
189	Samoa	WSM	26.172	15.3	Lower middle income
192	Congo, Dem. Rep.	COD	42.394	2.2	Low income
193	Zambia	ZMB	40.471	15.4	Lower middle income
194	Zimbabwe	ZWE	35.715	18.5	Low income

64 rows × 5 columns

In [41]:
```
df['IncomeGroup'] =='Upper middle income'
```

Out[41]:
```
0      False
1      False
2       True
3       True
4      False
       ...
190    False
191     True
192    False
193    False
194    False
Name: IncomeGroup, Length: 195, dtype: bool
```

In [42]:
```
df[df['IncomeGroup'] =='Upper middle income']
```

Out[42]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
2	Angola	AGO	45.985	19.100000	Upper middle income
3	Albania	ALB	12.877	57.200000	Upper middle income
10	Azerbaijan	AZE	18.300	58.700000	Upper middle income
16	Bulgaria	BGR	9.200	53.061500	Upper middle income
19	Bosnia and Herzegovina	BIH	9.062	57.790000	Upper middle income
20	Belarus	BLR	12.500	54.170000	Upper middle income
21	Belize	BLZ	23.092	33.600000	Upper middle income
24	Brazil	BRA	14.931	51.040000	Upper middle income
28	Botswana	BWA	25.267	15.000000	Upper middle income
33	China	CHN	12.100	45.800000	Upper middle income
37	Colombia	COL	16.076	51.700000	Upper middle income
40	Costa Rica	CRI	15.022	45.960000	Upper middle income
41	Cuba	CUB	10.400	27.930000	Upper middle income
48	Dominican Republic	DOM	21.198	45.900000	Upper middle income
49	Algeria	DZA	24.738	16.500000	Upper middle income
50	Ecuador	ECU	21.070	40.353684	Upper middle income
57	Fiji	FJI	20.463	37.100000	Upper middle income
60	Gabon	GAB	30.555	9.200000	Upper middle income
69	Grenada	GRD	19.334	35.000000	Upper middle income
82	Iran, Islamic Rep.	IRN	17.900	29.950000	Upper middle income
83	Iraq	IRQ	31.093	9.200000	Upper middle income
87	Jamaica	JAM	13.540	37.100000	Upper middle income

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
88	Jordan	JOR	27.046	41.000000	Upper middle income
90	Kazakhstan	KAZ	22.730	54.000000	Upper middle income
98	Lebanon	LBN	13.426	70.500000	Upper middle income
100	Libya	LBY	21.425	16.500000	Upper middle income
101	St. Lucia	LCA	15.430	46.200000	Upper middle income
112	Maldives	MDV	21.447	44.100000	Upper middle income
113	Mexico	MEX	19.104	43.460000	Upper middle income
114	Macedonia, FYR	MKD	11.222	65.240000	Upper middle income
118	Montenegro	MNE	11.616	60.310000	Upper middle income
119	Mongolia	MNG	24.275	20.000000	Upper middle income
122	Mauritius	MUS	10.900	39.000000	Upper middle income
124	Malaysia	MYS	16.805	66.970000	Upper middle income
125	Namibia	NAM	29.937	13.900000	Upper middle income
136	Panama	PAN	19.680	44.030000	Upper middle income
137	Peru	PER	20.198	39.200000	Upper middle income
143	Paraguay	PRY	21.588	36.900000	Upper middle income
146	Romania	ROU	8.800	49.764500	Upper middle income
157	Serbia	SRB	9.200	51.500000	Upper middle income
160	Suriname	SUR	18.455	37.400000	Upper middle income
169	Thailand	THA	11.041	28.940000	Upper middle income
171	Turkmenistan	TKM	21.322	9.600000	Upper middle income
173	Tonga	TON	25.409	35.000000	Upper middle income

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
175	Tunisia	TUN	19.800	43.800000	Upper middle income
176	Turkey	TUR	16.836	46.250000	Upper middle income
183	St. Vincent and the Grenadines	VCT	16.306	52.000000	Upper middle income
191	South Africa	ZAF	20.850	46.500000	Upper middle income

In [48]:
```
df['suresh'] = df.BirthRate + df.InternetUsers
df.head()
```

Out[48]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup	suresh
0	Aruba	ABW	10.244	78.9	High income	89.144
1	Afghanistan	AFG	35.253	5.9	Low income	41.153
2	Angola	AGO	45.985	19.1	Upper middle income	65.085
3	Albania	ALB	12.877	57.2	Upper middle income	70.077
4	United Arab Emirates	ARE	11.044	88.0	High income	99.044

In [50]:
```
df = df.drop('suresh',axis=1)
df.head()
```

Out[50]:

	CountryName	CountryCode	BirthRate	InternetUsers	IncomeGroup
0	Aruba	ABW	10.244	78.9	High income
1	Afghanistan	AFG	35.253	5.9	Low income
2	Angola	AGO	45.985	19.1	Upper middle income
3	Albania	ALB	12.877	57.2	Upper middle income
4	United Arab Emirates	ARE	11.044	88.0	High income

In [ ]:

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